Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **Python** services, **integrations**, and workflow automations on **Azure** and AWS with SQL, monitoring, and row-level data security, plus a **healthcare** records app, using AI-assisted development daily.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*
- Built **Python** FastAPI services and integrations across Azure SQL, SQL Server, PostgreSQL, and Blob Storage with validation, error handling, and reconciliation, running 5K+ monthly jobs at 99%+ reliability.
- Secured sensitive multi-tenant data with Azure SQL **row-level security** for 500+ users and troubleshot slow queries, cutting dashboard latency from 1.2s to 350ms through indexing and access-pattern tuning.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*
- Built a Python and Flask application with React views that automates grading through GPT-4o, Claude, and Qwen-VL with OCR, serving 150+ instructors and cutting feedback turnaround 60% via **workflow automation**.
- Implemented secure PDF, DOCX, and ZIP ingestion with concurrent batches of 1K+ files on AWS RDS, with timeout fallback and per-call usage logging, shipping through GitHub Actions **CI/CD**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*
- Automated cross-platform data-sharing **workflows** on a microservice architecture that cut manual data entry 35%, working with Microsoft and GitHub stakeholders from requirement analysis through deployment and production support.
- Optimized large-scale **API integrations** with caching and pagination, improving response times 30%, and automated nightly Jenkins test runs that emailed failure logs and coverage reports for fast troubleshooting.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*
- Built eight microservices on **Azure** App Services and Azure SQL for authentication, role-based access, billing, and email and SMS notifications, syncing 2M+ records across locations in real time.
- Ran production **monitoring** with Sentry, Azure Monitor, and Kibana, encrypted Azure SQL data behind firewalls and RBAC, and used Google Analytics insights to cut resource wastage 42%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*
- Built Python Flask APIs integrating existing AWS-hosted auth microservices, Firebase, and third-party payment providers, mapping 12+ **integration points** with the client and sustaining sub-200ms latency across 50K+ records.
- Automated refund, dispute, and chargeback handling through idempotent payment **webhooks**, cutting manual work 70%, and used Firebase transactional locking to prevent double-booked time slots under concurrent bookings.

Skills

Languages: Python, SQL, TypeScript, JavaScript, Java
Backend & Integrations: FastAPI, Flask, Node.js, REST APIs, Webhooks, Microservices, Health-Record APIs (ABDM), Twilio
Cloud & Data: Microsoft Azure, Azure SQL, Blob Storage, AWS (S3, RDS, Glue), SQL Server, PostgreSQL, MongoDB, ETL Pipelines
Reliability & Security: Monitoring, Alerting, Sentry, Azure Monitor, Error Handling, Row-Level Security, JWT, RBAC, Snyk
Tooling & AI: Git, GitHub Actions, Jenkins, Docker, CI/CD, Claude Code, LLM Integration
Practices: Technical Documentation, Stakeholder Communication, Requirements Gathering, Agile

Projects

Microsoft Graph Workflow Automation | *Python, Azure Functions, Microsoft Graph API, SharePoint, Power BI*
- Designed a Python Azure Functions service using **Microsoft Graph API** to triage shared-mailbox requests into SharePoint lists and Teams alerts, redact PHI from logs, and feed a **Power BI** dashboard.

HMIS – Hospital Management Information System Live | *TypeScript, MERN, ABDM APIs, Twilio, Sentry*
- Won Smart India Hackathon with a **healthcare** app integrating ABDM health-record APIs behind patient consent, automating appointment workflows and Twilio reminders, and cutting administrative workload 30% across 100+ daily appointments.

Chat2SQL Code | *Java, TypeScript, SQL Server, PostgreSQL, Docker*
- Extended an AI-assisted SQL workspace on Java and TypeScript supporting 10+ databases including SQL Server, generating schema-aware queries at 82% accuracy for faster **data troubleshooting** and exploration.